ATRC ADVANCED TECHNOLOGY PIONEERS 2026 · ENEC TRACK

# QANDEEL

## قنديل

إدارة أسراب قناديل البحر وإعادتها إلى البحر حيّة

## Herd, don’t harvest: autonomous, live-release management of Blue Blubber jellyfish swarms after ENEC’s early warning


**Team overview · Team Qandeel**

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EXECUTIVE SUMMARY

# We keep swarms out of the intake and return them to the sea alive

**Problem**

Blue Blubber swarms (30–45 cm bells) drift onto seawater intakes. Clearing them takes heavy manual work and almost none survive.

**Solution**

After ENEC's alert, a drift planner decides when to act; a bubble curtain at the breakwater gap holds the swarm out; two uncrewed boats tow a boom with a closed-bottom bag and release the jellyfish alive. An AI gap camera is the backup.

**Evidence**

Simulation on real 2025 Gulf currents and wind, stress-tested against physics the planner does not know; online dashboard; 25 tests; open code.

**Next**

Recalibrate the model with ENEC's alert history, screen logs, flow and depth data, then a one-section sea pilot.

## ~3% vs ~23%

of arriving jellyfish reach the gap before the curtain runs: Qandeel vs a daylight camera alone (stress test on real 2025 data; mean of 3 × 100 alerts, range 2.7–3.7% vs 20–26%)

## 1–2 days

of warning for the boats and the curtain, for ~12 extra curtain hours per alert vs camera-only (~USD 500 of extra energy)

## 84–100%

of jellyfish kept by the boom with our retention bag (model), vs 0–55% for an open oil-spill boom

## 29–46%

of the energy of a fixed airflow, by setting air hour by hour from real currents and wind (variable-speed compressor)

THE PROBLEM AND WHY IT MATTERS

# A large, local, hard-to-handle jellyfish

ENEC already detects swarms offshore. The open question is what to do next: manage high volumes with less manual work, less harm and low energy, in line with the UAE Net Zero 2050 and biodiversity goals.

### The target: Blue Blubber

- One of the two dominant jellyfish on the UAE coast in 2019–2022 surveys (Aravind 2024, UAEU)
- 30–45 cm bell (ENEC brief; SeaLifeBase max ~35 cm); no published weight: ~1–3 kg across 30–45 cm by size scaling; we use 2 kg
- Lives near the surface of bays and estuaries, ~0–1 m (SeaLifeBase)
- Small adults swim ~4 cm/s (lab data); large adults unmeasured, so we test 10 and 20 cm/s

### What happens at intakes

- Swarms follow the intake flow and pile on the screens, cutting cooling water
- Worldwide this has forced power reductions and shutdowns (e.g. Torness 2011, Oskarshamn 2013) (press reports)
- In the region, ~300 t of jellyfish clogged intakes in Oman (Gulf News, Ghubra); in Kuwait, Doha West's intake has had to close for hours to days (Neelamani & Al-Hulail 2010, KISR)

### Options today, compared

| OPTION | JELLYFISH | DRAWBACK |
|---|---|---|
| Rake screens (now) | die | reactive, heavy labour, outage risk |
| Fixed nets offshore | die, bycatch | clog in big blooms |
| Cutting robot (JEROS) | shredded | ~0.9 t/h, kills |
| Flexible screens across the channel mouth (KISR, Kuwait) | kept out alive | clog; regular cleaning; costly |
| Harvest for food | used | no UAE market yet |
| **Qandeel** | **released alive** | unproven at sea: pilot first |

We found no published management trial on this species.

OUR SOLUTION · SYSTEM DESIGN

# Guard the mouth, move the water, keep them alive

**ENEC alert**
swarm position
(existing system)

→

**Qandeel planner**
- chance of arrival and window
- curtain switch-on time
- airflow, crew side, release point
- live current and wind forecast

→

**Bubble curtain**
at the intake mouth,
only when needed

→

**Uncrewed boats + boom + bag**
gather in the water

→

**Live release**
down-current

AI gap camera + pressure sensors feed back




### 1 · The mouth is a chokepoint

Seawater reaches the pumps through one ~3 km channel; its mouth measures 967 m on satellite images.

### 2 · Move the water, not the animal

A rising-bubble current holds jellyfish out at sea. Nothing is cut or netted, and unlike jets or pumps nothing moves underwater: the compressor is on land. Ten ~97 m sections; only those in the swarm's path run.

### 3 · Proven tools, made autonomous

Oil-spill booms on uncrewed boats, plus a bag that stops leaks. Release west, away from the discharge.

# One swarm, from alert to live release

1. **Alert.**
   ENEC reports a swarm, e.g. 15 km NNW. The planner receives it automatically.
2. **Plan.**
   400 simulated drifts of 60 swimming jellyfish each, on the live current and wind forecast: 44% chance of arrival, window 16–73 h.
3. **Prepare.**
   Curtain on at +13 h; uncrewed boats staged on the approach side; release point chosen.
4. **Deflect.**
   Curtain across the gap holds the swarm outside; airflow follows the current.
5. **Gather.**
   Two uncrewed surface vessels, supervised from shore, close a U-boom with a closed-bottom bag.
6. **Release.**
   Tow ~3 km NNW (≤0.2 m/s through the water: ~7 h over the ground, up to a day against strong inflow) and open the bag; ~60% drift back and are held again.
7. **Learn.**
   An AI camera counts jellyfish at the gap and starts the curtain if they arrive early. Its records are the training data for a learning model that corrects the drift forecast.

**44%**
chance the swarm reaches the gap

**16–73 h**
arrival window after the alert

**+13 h**
curtain switch-on

**3 km NNW**
release point (~7 h tow)



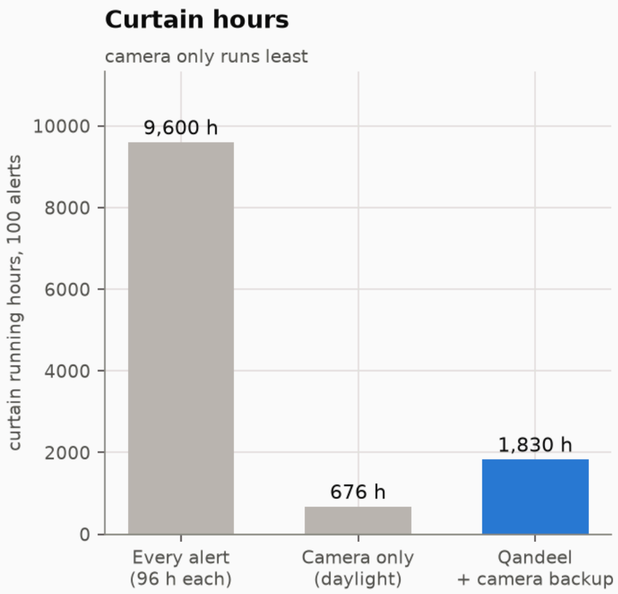

From the live dashboard, alert 15 km NNW: swarm cloud, release point, curtain status. Data: Copernicus SMOC + ERA5, summer 2025.

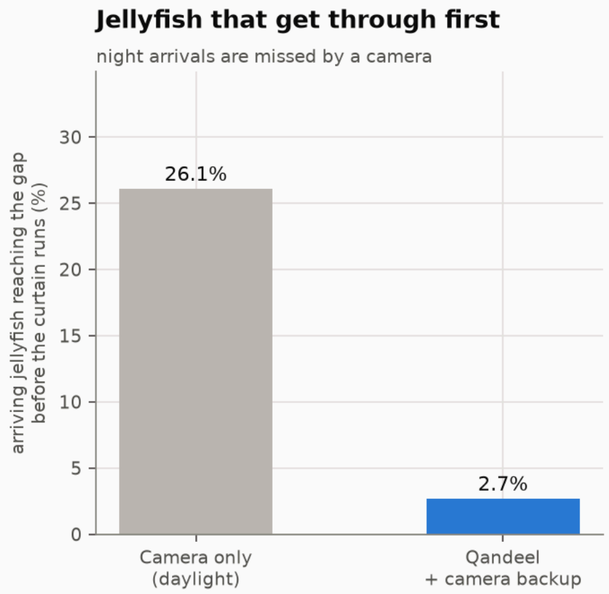
### How the plan is calculated

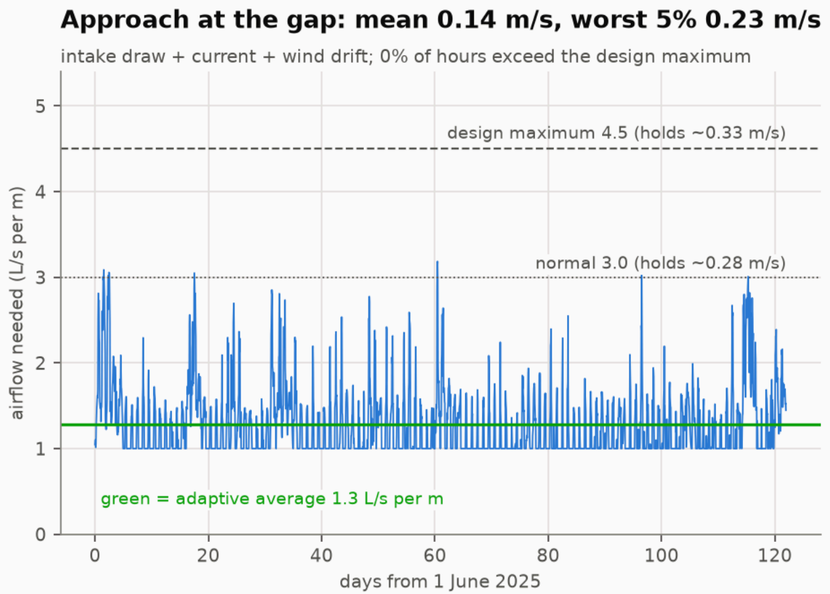
**Arrival:** 400 runs × 60 virtual jellyfish moved every 10 min by current, 1% wind drift, swimming and mixing; chance = runs where 5% come within 2 km.
**Switch-on:** if chance ≥10%, earliest likely arrival − 3 h; off once the window has passed and the camera sees none for 6 h.
**Release:** of 50+ points west to NNE, the one removing most jellyfish per boat-hour, avoiding strandings near the plant.
**Airflow:** set hourly to beat the approach current with a margin.

# Better protection than a camera alone, at a known cost

**Stress test, real 2025 season: Qandeel lets ~3% through vs ~23% for a camera alone (mean of 3 runs; range 2.7-3.7% vs 20-26%)**




100 alerts at random times and places in the real 2025 season; the truth uses physics the planner does not know. Base case 1% wind drift. “Arrived” = within 2 km of the gap (15 km NNW alert: 44%; within 500 m: 31%). Bars show one run (2.7% vs 26.1%); the card gives the mean of three.

### ~3% vs ~23% get through

Truth: currents ×0.7–1.3, wind drift ×0.5–1.7, plus a 0–3 cm/s current the planner doesn’t know. A daylight camera misses night arrivals.

### The cost

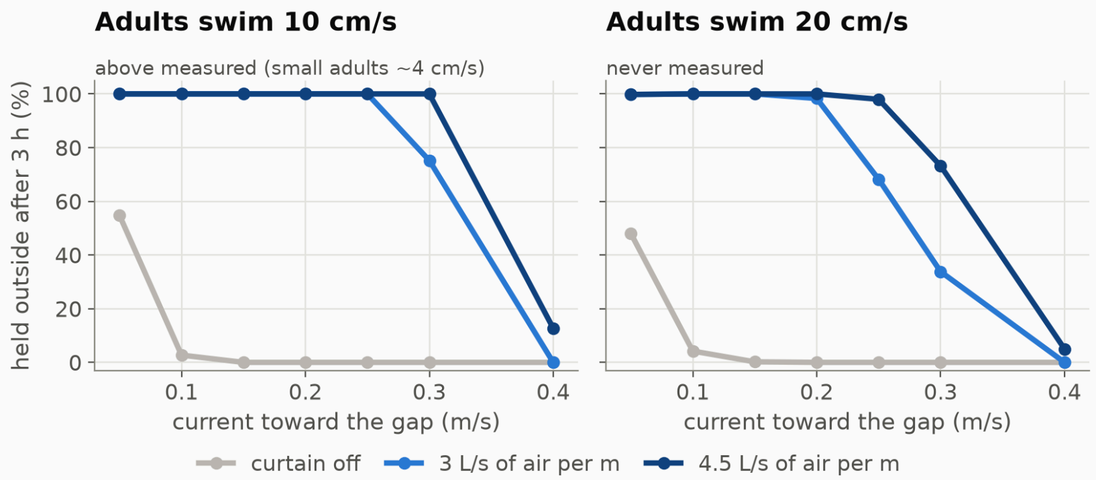
~1,800 curtain hours vs ~700 camera-only: ~12 h and ~USD 500 more energy per alert; ~80% less than running on every alert. In return: 1–2 days of warning.

### What this is not

Still simulation against simulation, not real jellyfish. Next: compare drift with real Gulf drifter tracks and ENEC swarm records.

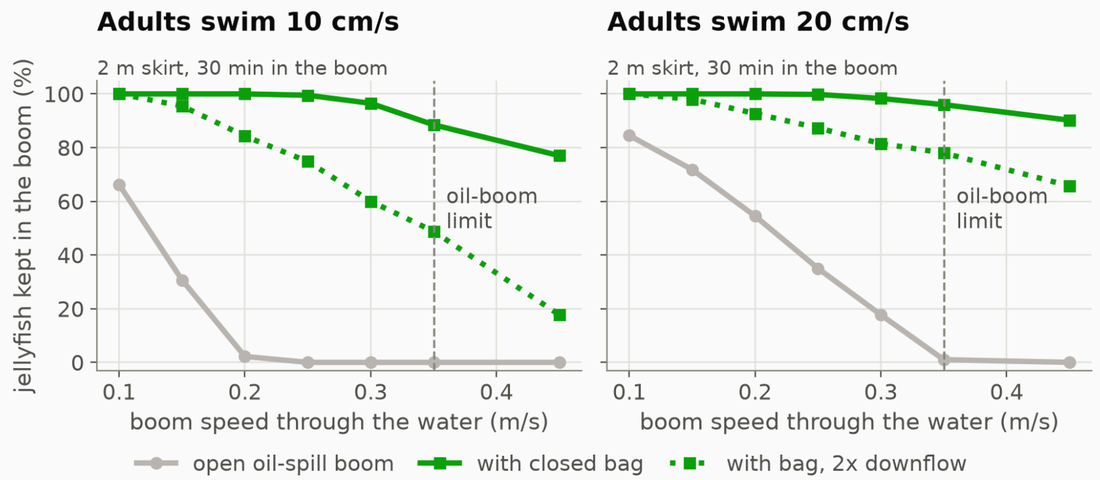
# Oil-spill booms leak jellyfish; a closed bag fixes it

## More air holds against faster water




Each point: 400 simulated jellyfish. Bubble-driven surface current from the standard bubble-curtain formula (Bulson 1968: current grows with the cube root of airflow); its constant is confirmed in the pilot.

## Open booms leak jellyfish; the bag keeps them




A towed boom pulls water down behind it (“downflow”): we assume 25% of tow speed, dotted line 50%. Jellyfish depths from field data.

### Curtain holds, if adults swim slowly

Water approaches the gap at 0.14 m/s on average, 0.23 m/s in the worst 5% of hours (0.20 / 0.36 if wind drift is 3%). 3 L/s per m holds ~0.27 m/s; 4.5 holds ~0.33. Over a full 82 h event: 100% held at 0.2 m/s, but 78% if adults swim 20 cm/s.

### Why the bag matters

Oil floats back up; jellyfish drift with the water and are dragged under the skirt well below the 0.35 m/s oil-boom limit (2–55% kept at 0.2 m/s). A closed-bottom fabric bag keeps 84–100% in the model.

**Where held jellyfish go.** Boats clear only ~0.4–1.4 t/h, so a 300 t bloom (Oman) would take them 9–30 days: they don’t remove the swarm, they clear what piles up at the mouth. The curtain holds the swarm while the sea moves it on: in our 2025 data the flow runs along the coast at ~0.07 m/s (~6 km a day) and turns offshore 22% of the time (onshore spells last a median 14 h). The curtain stays on until the planner’s window has passed and the camera has seen no jellyfish at the mouth for 6 h, so a held swarm cannot pour in when it stops.

# Gentle on marine life, built from parts that exist today

## Marine-life protection

| RISK | HOW WE REDUCE IT |
|---|---|
| Bubbles drawn into the intake | Curtain on the seaward side; bubbles surface within a minute, in the outer basin; bubble detector on the intake line; curtain moved out if the pilot shows carry-over |
| Nuclear approval, security | FANR approval from day one; kit stays outside safety-related intake structures; boats run by plant staff under security-zone rules |
| Jellyfish harmed | Nothing cuts, pumps or nets them: the curtain's surface current pushes them back. Effect of bubbles on the bell not yet known: checked in the pilot |
| Turtles, dolphins, dugongs, seagrass | Escape gap, slow tow, camera stop on sighting; pipe routed away from seagrass and coral; release points kept out of protected areas |
| Crowding in the bag during the tow | Bag stays underwater, where jellyfish are nearly weightless, and seawater flows through the fabric: we expect far less crushing than in hauled nets. At most half full; opened at sea by 12 h. Survival is a pilot pass mark |
| Released jellyfish strand | Left alone, 55% of this swarm strands within 4 days; released 3 km out, 29% do, most still within 10 km of the plant. Release points refined with EAD |
| Sediment, bubbly water | Diffuser pipe on legs 0.5–1 m above the bed, so sand cannot bury the holes and block the air (less repair work); turbidity watched; no boats or divers in the bubble zone while it runs |

## Feasibility

| ITEM | ESTIMATE (967 M MOUTH) |
|---|---|
| Compressed air | 2.9 $m^3/s$, ~440 kW for the 967 m mouth, oil-free (ISO 8573-1 Class 0) |
| Capital (indicative) | USD 1.2–3.5 M: 967 m curtain, variable-speed compressor + standby, boom, bags, two uncrewed boats, sensors |
| Running cost | USD 0.18–0.45 M a year (operators, boat upkeep, diver cleaning, servicing). A full 72 h event: ~31 MWh, ~USD 3,100; 12.6 t $CO_2$ on a gas grid, 0.4 t on the plant's own power |
| Value | Torness (UK) lost an estimated ~USD 1.5 M (~£1 M) a day while jellyfish kept it shut in 2011 (press estimate): avoiding 0.8–2.3 such days repays the capital |
| Capacity | Gathering 4–43 t/h, but the loaded tow sets the pace: ~0.4 t/h per boom unit, ~1.4 t/h with three uncrewed tugs swapping bags. The curtain does the bulk work; boats clear what builds up |
| Site works | Compressor house on land at the breakwater root; air main with power and fibre along the ~3 km breakwater; valve chamber and camera mast at the tip; boat base inside the channel |
| Gulf hardening | Membrane diffusers, foul-release coating, modular sections, lifted out off-season |

ASSUMPTIONS (NO DATASET PROVIDED) AND LIMITATIONS

# What we assume, and what we measure first

| ASSUMPTION | BASIS | IF WRONG |
|---|---|---|
| Intake mouth 967 m wide (measured), ~9.5 m deep | Satellite images; public channel depth | Depth to survey; a shorter curtain inside the channel would cut air and cost |
| Alert gives position, hours of notice | Brief: “time to prepare” | Camera trigger switches the curtain on |
| Swarm in top 1 m; adults swim ≤10–20 cm/s | Field and lab studies (Australia) | Deeper skirt; tested at 20 cm/s |
| Swarm density 0.1–1 per $m^3$ | No published figure | Capacity scales linearly |
| Extra wind drift 1% of wind (0–3%) | Ocean model already has wind-driven drift | 28 km N: 3–29% arrival. Curtain sized for 1%; at 3%, ~8% of hours exceed its max (flagged) |
| Apex downflow 25–50% of tow speed; bag passes 20% | Model hypothesis | Measured in the sea pilot; bag tested before scale-up |

**Limitations:** tested only against simulations; currents from one model point, rounded to ~0.03 m/s by the free API (direct Copernicus download ready); no depth: the curtain draws water in near the bed, so deep or night-time jellyfish could pass under; species at Barakah and adult weight to confirm. **Why not harvest?** A full bag can go to a licensed processor; release is the default while no UAE market exists.

## What moves the results most

One unknown changed at a time, the rest at base values:

| UNKNOWN (RANGE) | RESULT |
|---|---|
| Current toward the gap (0.1→0.3 m/s) | held 100% → 76% |
| Downflow behind the boom (15→50%) | kept 100% → 88% |
| Bag fabric leak (10→40%) | kept 100% → 88% |
| Swim speed (5→20 cm/s) | held 100% → 99% |
| Gap width (500→1,200 m) | 226 → 541 kW (base 436) |
| Water depth (6→12 m) | 361 → 484 kW |

**So the pilot measures current, downflow and bag leak first.** Chart in the GitHub README.

HOW WE WOULD TEST AND DEMONSTRATE IT IF SHORTLISTED

# From simulation to sea, one measured step at a time

Done

### 1 · Simulation

Real-data planner, curtain, boom, release and logistics models; online dashboard; stress test; 25 tests.

Months 1–2

### 2 · ENEC data

Alert history, screen-clearing logs, intake flow, depth survey and species ID: recalibrate the drift, curtain and release models on the real site.

With ENEC

### 3 · Sea pilot

One curtain section and one uncrewed boom trial in bloom season; FANR approval; drift checked against tracked swarms; survival and effects on other species.

Then

### 4 · Scale-up

Full gap; live ENEC alerts into the planner; learning model trained on camera and swarm records corrects the forecast.

## Pass marks

- Pilot: ≥95% of jellyfish held at the curtain section at the measured current
- Pilot: bag keeps ≥90% at 0.2 m/s; ≥90% still alive 24 h after release
- No harm to non-target species; fewer screen-clearing hours

## Open work and AI use

Live dashboard: enechackathon.streamlit.app · code, data, 25 tests: github.com/Shaaha-7/ENEC. AI assistants helped search literature, check claims, draft text and write code; an open text-to-speech model voices the video. The team reviewed every decision and figure.

**Sources:** Aravind 2024 (UAEU) · SeaLifeBase · J. Exp. Biol. 2018 · FRDC 1999/138 · Bulson 1961, 1968 · UVM Sea Grant · Copernicus SMOC, ERA5 · Gulf News, Nat. Geo. 2013, Vice 2021 · IEEE Spectrum · Neelamani & Al-Hulail, ICHE 2010 (KISR) · Lo 1991 (bubble barriers) · Hydrotechnik Lübeck. Links, full tables and charts: GitHub README.

### Approvals we plan for

FANR for any change at the intake; EAD for marine works and moving live animals; Coast Guard and plant security for boats; an engineering safety case for air near the pumps.

### What we need from ENEC

Screen-clearing logs and alert history; intake flow rate; a depth survey of the channel mouth; species ID at Barakah; event frequency.

### Beyond the intake

Survey polyps on the breakwater and intake structures, where blooms start. The same kit fits Gulf desalination plants (Taweelah, Fujairah, Oman) and other coastal power stations.